Alan Nguyen Pham

Mechanical R&D | Soft Robotics | Physical Interaction

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Mechanical engineering Ph.D. candidate building soft robotic materials, reconfigurable mechanisms, and motion prototypes for human-facing physical systems. Targeting mechanical R&D and Imagineering-aligned roles where CAD/fabrication, experimental characterization, pneumatic/compliant mechanisms, and visible physical behavior meet.

Research Focus

Soft robotics; morphing mechanisms; robotic materials; compliant mechanisms; pneumatic actuation; mechanical design; CAD/fabrication; force/stiffness testing; motion systems; haptics; human-robot interaction; physical HCI; experimental mechanics; nonlinear dynamics; shape displays; adaptive interfaces.

Education

Doctor of Philosophy, Mechanical Engineering — **Expected 2027**
Worcester Polytechnic Institute, Worcester, MA — Advisors: Dr. Pratap Rao and Dr. Cagdas Onal

Master of Science, Mechanical Engineering — **2022**
California State University, Long Beach, Long Beach, CA — Advisor: Dr. Allen Teagle-Hernandez

Bachelor of Science, Mechanical Engineering — **2019**
California State University, Maritime Academy, Vallejo, CA — Advisor: Dr. Tomas Oppenheim

Awards

NSF NRT Future of Robots in the Workplace – Research and Development (FORW-RD) Fellowship — 2024
WPI Intercollegiate Soft Robotics Competition – 1st Place — 2024

Selected Public Work

- Sarrus: programmable soft-robotic cells for shape change, stiffness control, object manipulation, locomotion, and visible motion behavior.
- AO Labs: deployed portfolio of research projects, prototypes, tools, and public work connecting mechanical systems, physical interaction, and autonomous software.

Research Experience

Worcester Polytechnic Institute, Worcester, MA — **Aug 2023 – Present**
Graduate Researcher – NanoEnergy Lab and Soft Robotics Lab — *Advisors: Dr. Pratap Rao and Dr. Cagdas Onal*

- Led first-author research on reconfigurable Sarrus-linkage soft-robotic cells that turn one monolithically printed pneumatic Sarrus/PneuNet unit into planar, cylindrical, and locomotion assemblies.
- Designed, fabricated, and characterized 2-by-2 pneumatic modules with cross-cell channels, snap-fit connectivity, and built-in bending bias for repeatable out-of-plane morphing.
- Assembled programmable morphing-surface arrays demonstrating single- and double-dome shape display, bilayer overhangs, tennis-ball transport/capture, and horizontal, diagonal, and circular traveling-wave deformation.
- Reassembled modules as cylindrical soft robots for continuum bending, grasping/placement, a 200 mm peristaltic crawl, and self-rolling locomotion.
- Quantified pressure-dependent expansion, bending angle, force-displacement response, hysteresis, and variable stiffness across cell, module, and array scales to connect geometry and actuation pattern to task-level behavior.

NASA L' SPACE (NPWEE), Tempe, AZ — **Aug 2025 – Present**
Soft Robotics Graduate Researcher – Project SELENE

- Advanced Project SELENE, a multidisciplinary exploration-robotics effort developing adaptable, resilient robotic systems for extreme and remote environments, including beyond-Earth mission concepts.
- Developed a modular cable-driven soft robotic arm concept for terrain navigation and regolith sample collection in hard-to-reach exploration settings.

- Refined the "Spirobs" arm design by integrating triple-cable actuation with an elastic braided-steel backbone for multidirectional bending, mechanical compliance, dust-resistant operation, and reduced maintenance.
- Led CAD updates, fabrication planning, 3D-printing decisions, assembly strategy, and structured test/evaluation planning while guiding undergraduate teammates and coordinating with NASA subject matter experts.

Sandia National Laboratories, Albuquerque, NM **June 2025 – Aug 2025**
Graduate Research Intern – Nonlinear Mechanics and Dynamics (NOMAD) Research Institute

- Co-developed a NOMAD parameter study quantifying how finite-element setup choices affect nonlinear-dynamics predictions for an electromechanical ratcheting mechanism with intermittent contact, friction, springs, and pawl-gear interactions.
- Built comparable analytical, MATLAB multibody, and finite-element submodels for pin-spring-pawl, pin-pawl, and pawl-gear interactions under haversine shock and sinusoidal vibration.
- Ran structured sweeps across momentum-balance iterations (1–100), mesh density, and processor count (50–250%) to separate physical mechanism response from solver and modeling sensitivity.
- Identified sensitivity regimes in which pin-spring-pawl models converged while pin-pawl and pawl-gear cases showed contact/friction divergence, mesh-related lock-up, and environment-dependent sensitivity.
- Presented final results showing that simulation sensitivity appears at submodel scale, informing more defensible setup choices before full assembly-level ratcheting studies.

Worcester Polytechnic Institute, Worcester, MA **Aug 2023 – Present**
NSF NRT FORW-RD Fellow and Trainee

- Selected as an NSF NRT FORW-RD trainee in an interdisciplinary program on robotic interfaces and assistants for future workplaces spanning robotics, mechanical engineering, computer science, materials, business, humanities, and social sciences.
- Connected doctoral research to haptic and virtual training, soft robotic teleoperation, compliant interaction, and human guidance of robots for high-skill physical tasks.
- Developed broader-impact framing for robotics and AI systems that operate near people, emphasizing workplace adoption, trust, physical interaction, training, and responsible deployment.
- Secured a $1,000 grant supporting equipment, travel, and professional-development costs.

Worcester Polytechnic Institute, Worcester, MA **Aug 2023 – Dec 2023**
Research Assistant – Cognitive Medical Technology Laboratory

- Fabricated and assembled low-cost open-source actuation hardware for concentric tube continuum robots, a miniaturized robot class for precise navigation in constrained spaces and minimally invasive surgery.
- Contributed to an accessible CTR research platform with easy-to-source components, bills of materials, assembly instructions, and guided learning materials for early graduate students.
- Integrated actuation hardware, MATLAB forward kinematics, and optical-tracking experiments to validate end-effector displacement.
- Coordinated sourcing, fabrication, and assembly across WPI's Cognitive Medical Technology Lab, PracticePoint, and Innovation Studio for two complete CTR systems.

California State University, Long Beach, Long Beach, CA **Sept 2022 – Aug 2023**
Research Assistant – Multiscale Mechanics of Materials Group *Advisor: Dr. Mortaza Saeidi-Javash*

- Built a literature foundation for a flexible-electronics research program covering printed and conformal sensors, 2D nanomaterial inks, and wearable human-machine-interface devices.
- Synthesized fabrication approaches across inkjet, screen, aerosol-jet, extrusion, and direct-ink-write printing using graphene, MXene, MoS_2, thermoelectric, and piezoelectric material systems.
- Evaluated sensitivity, selectivity, response time, mechanical compliance, and durability of flexible physical, chemical, and biosensors for soft robotics, human-robot interfaces, structural health monitoring, and wearable systems.
- Collaborated with graduate researchers on a direct-ink-write temperature sensor printed on a flexible PET substrate.

California State University, Long Beach, Long Beach, CA **April 2021 – July 2022**
Graduate Researcher – Violin Acoustics and Structural Dynamics *Advisor: Dr. Allen Teagle-Hernandez*

- Completed a master's thesis translating perceived violin quality into measurable vibroacoustic features, linking structural design, harmonic balance, frequency response, and performer/listener perception.

- Designed a controlled comparison of high- and low-cost modern violins, standardizing strings and fittings while documenting top plate, bridge, f-hole, bass bar, sound post, and tailpiece variables.
- Built a LabVIEW/NI data-acquisition workflow with PCB accelerometers, an array microphone, and an impulse modal hammer; collected 67-point top-plate maps and anechoic-chamber acoustic measurements.
- Linked FFT-derived violin response features to modal balance and perceived tone.

California State University, Maritime Academy, Vallejo, CA **Sept 2018 – May 2019**
Undergraduate Researcher – Cross-Flow Tidal Turbine *Advisor: Dr. Tomas Oppenheim*

- Conceptualized a Darrieus cross-flow tidal turbine, defined rotor geometry and build constraints, and selected 3D-printed blades with fiberglass coating as the manufacturing approach.
- Verified performance through water-tunnel power assessment and ANSYS structural analysis, evaluating blade and hub stresses under hydrodynamic loading and documenting operation around 25 rpm.
- Fabricated a full-scale prototype using additive manufacturing and sheet-metal forming informed by scaled water-tunnel experiments.

Publications and Research Outputs

- Pham, A. N., Rao, P., & Onal, C. D. (2026). Manuscript in prep.: *Reconfigurable Sarrus-linkage cells for multifunctional soft robots*.
- Bonofiglio, K. (2023). *Making Concentric Tube Robots More Accessible: An Open-Source Platform with Easy-to-Source Components and Guided Learning Materials* [Master's thesis, Worcester Polytechnic Institute]. Research contributor to open-source CTR hardware assembly and validation support.
- Pham, A., & Teagle Hernandez, A. (2022). *Structural Properties and Perceived Acoustic Quality of Modern Violins* (Publication No. 29065875) [Master's thesis, California State University, Long Beach]. ProQuest Dissertations and Theses Global.

Professional Experience

Worcester Polytechnic Institute, Worcester, MA **Aug 2023 – Present**
Teaching Assistant

- Developed learning activities for Stress Analysis, Statics, and Computer Aided Design with professors and teaching teams.
- Led weekly conference sessions and office hours for 90+ student groups, translating difficult engineering material into interactive problem-solving sessions.
- Designed hands-on learning sessions and guided students in connecting classroom mechanics concepts to physical models and engineering judgment.

NLS Lighting, Carson, CA **April 2022 – July 2023**
Mechanical Engineer

- Led design-to-release luminaire product improvements across thermal testing, IP testing, FEA static simulation, fabrication, assembly, and installation support.
- Produced complete product design packages, including 2D drawings, bills of materials, engineering change orders, installation instructions, and application documentation.
- Collaborated with Washington, DC stakeholders to develop a retrofit kit for upgrading more than 75,000 street and alley lights to energy-efficient LEDs as part of a multi-million-dollar Smart Street Lighting project.

Hubbell, City of Industry, CA **May 2021 – Aug 2021**
Mechanical Engineer Intern

- Managed fixture assembly and DLC V5.1 certification planning, reducing test cost and labor while accounting for SKU quantities and sales volume.
- Performed impact testing and FEA validation on luminaire assemblies, checking bracket, lens, and LED-PCB designs in Solid Edge.
- Supported launch of the Prevent vandal-resistant luminaire series through baseline testing, part sourcing, and assembly coordination with new product design teams.

Academy Inc., Montebello, CA **July 2019 – Aug 2020**
Mechanical Design Engineer

- Implemented parametric product and pricing configurations, reducing design and quoting time.
- Engineered shade sails, awnings, and tensioned fabric structures for manufacturability and field installation.
- Managed projects from site measurements and material specifications through coordination and scope execution.

American Metal Bearing Company, Garden Grove, CA **May 2019 – June 2019**
Mechanical Engineer Intern

- Optimized thrust, line-shaft, stern-tube, and strut bearing components using SolidWorks/Abaqus FEA, reducing material and labor cost.
- Coordinated manufacturing flow for line-shaft spherical bearings from rough casting through inspection, finishing, packaging, and shipment.
- Revised bearing-component drawings under Engineering Change Requests while maintaining GD&T standards.

Technical Skills

Mechanical design	SolidWorks, Onshape, PTC Creo, FEA review, GD&T-aware drawings, BOMs, ECOs, DFM, assembly planning, fixture design, prototype iteration.
Soft robotics	Pneumatic actuation, compliant mechanisms, continuum robots, modular soft robots, morphing surfaces, shape displays, haptics, human-robot interaction, robotic prototyping.
Software	CUBIT, Sierra/SolidMechanics, ParaView, MATLAB, Simulink Simscape Multibody, Abaqus, Ansys FEA, Python, C/C++, ROS2, LabVIEW, Minitab, MS Office.
Experiments	Modal testing, vibroacoustic measurement, FFT-based signal analysis, anechoic-chamber testing, optical tracking, pressure/force/displacement characterization, design of experiments.
Hardware	3D printing, laser cutting, soldering, Arduino, Raspberry Pi, National Instruments DAQ, strain gages, thermocouples, pressure sensors, force testing (Mecmesin), hand tools.

Relevant Coursework

Soft Robotics; Advanced Dynamics; Ethics and Communication in Robotics and AI Research; Foundations in Robotics; Advanced Mechanics of Deformable Bodies; Mechatronic System Design; Design of Experiments; Haptic Systems for Virtual Reality and Teleoperation; Engineering Vibrations; Automatic Feedback Control; Instrumentation and Measurement Systems.

Campus and Community Involvement

Member, American Society of Mechanical Engineers	March 2015 – Present
WPI Video Game Orchestra	Nov 2024 – Present
WPI Tennis Club	Aug 2024 – Present
WPI Philharmonic Orchestra, 1st Violinist	Nov 2023 – Present
CSULB Long Beach Lunabotics, Mechanical Engineer	Sept 2020 – May 2022
CSULB Beach Launch Team, Avionics Engineer	Sept 2020 – May 2022
CSULB Bob Cole Symphony Orchestra, 2nd Violinist	Sept 2019 – March 2020
Rebuilding Together Solano County, Volunteer	Aug 2015 – Aug 2016